

10.1 Evaporated Aluminium Release Solution

Purpose:

The aluminium evaporated in the machine has approximately a 70% collection efficiency onto the substrate being processed.

Unfortunately, the other 30% adheres which is not evaporated onto the substrate and finds its way onto the machine surfaces, around the evaporation source.

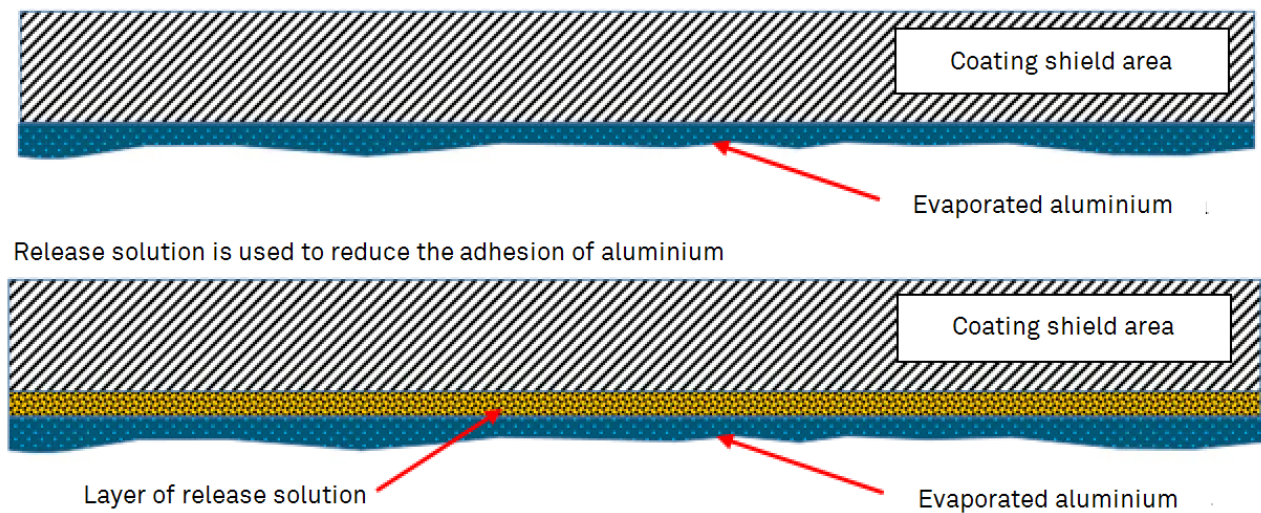


Figure 10.1-1 Sketches showing surface application benefits of release paint

This will save a considerable amount of time and physical effort in cleaning down these machine surfaces.

Surfaces such as the “drum edge shields” and the shutter will need to be cleaned and have the solution reapplied every process cycle.

Other surfaces, which are not in the direct evaporation “line of sight”, such as the viewing strobe, only need to have the solution applied when these parts are cleaned in a less regular regime.

The solution used is also used in other industries, such as “metal casting”, also used as a release coating to allow easier removal of the “moulds”.

Boron Nitride Water-Based Release Solution:

Supplier directions for use:

Shake the container vigorously and apply with a brush to give a uniform coating. If necessary, dilute with water or ethanol.

	Note !
	Do not allow release paint to come into contact with the evaporator clamping block contact area, as potentially, is can insulate the two electrical contact surfaces.

Available from all the evaporator manufacturers and other industrial sources.

Aquadag Water-Based Graphite Solution

Aquadag (trade name) is an alternative release solution. It is described as a colloidal dispersion of extremely fine, pure graphite in a water carrier that dries to form an adherent film on virtually all surfaces. It exhibits excellent parting abilities and an extremely low coefficient of friction.

Dilution:

Aquadag is supplied as a concentrate and may be diluted before use.

Dilution ratios vary according to end-use requirements. It is suggested to contact the supplier for further information.

	Note !
	Always add the water to the concentrate, never vice versa

Supplier directions for use:

Stir the concentrate thoroughly before diluting to break up the gel structure. Then, stirring gently, slowly add small amounts of water to the concentrate until a fluid consistency is obtained with no small gels. Stir in the remaining water to attain the desired dilution. Tightly reseal container after use.

<http://na.henkel-adhesives.com/product-search-1554.htm?nodeid=8798046388225>

Application

Aquadag may be applied using conventional spray, brush, or dip methods.

Curing

Aquadag will air dry to touch in 12-15 minutes, depending on room temperature and humidity.

10.2 Graphite Tape

The purpose of the Graphite tape is to provide good electrical contact between the thermal evaporators and the water-cooled clamping blocks.

Directions for Use

Cut into strips the same width as the clamping blocks. Use a single thickness to cover the inside faces the of the evaporator clamps before inserting the evaporator.

Availability

Available in rolls 30m long x 15mm wide x 0.2mm thick.

Available from all evaporator manufactures and other suppliers.



Figure 10.2-1 Graphite tape

Graphite Tape for Coating Shield Sine Waves

There is a requirement to place a strip of graphite tape behind the copper sine wave fingers when installing them onto the coating shield for heat transfer purposes.

Available in rolls 12m long x 40mm wide x 0.5mm thick.

Bobst Manchester Ltd purchase tape from Gee Graphite -

<http://geegraphite.com/Products/TabId/249/PID/2517/evl/0/CategoryID/33/CategoryName/Graphite-Foils/Default.aspx>

10.3 Ceramic Evaporators

There are a number of suppliers of thermal evaporators. When ordering new supplies refer to the evaporator information in the machine specification.

Bobst Manchester K5 Expert range of machines are normally designed to use:-

- 125mm long x 40mm wide x 12.5 mm deep evaporators

The size to be used in individual machines is specified in the technical specification.

The above size is a common to the metallizing industry and are available from all the suppliers.

The amount of evaporators required in a normal production environment will depend upon the deposition level during coating cycles. Normal life should be approximately 15 production hours.

The number of evaporators that are mounted in the machine is listed in the machine's technical specification.

The machine will automatically select only the required number of the evaporators that is proportional to the width of the material being coated. Therefore in the case of processing a roll of substrate, less than the machines maximum width, the number of evaporators that are heated up are reduced.



Figure 10.3-1 Typical ceramic evaporators

The machine will accept the latest various shaped evaporators from the suppliers.

A copy of the relevant Bobst Manchester Limited data sheet is attached for the user's information.

10.4 Aluminium Wire

Several different alloys of aluminium wire are available and used in the metallizing industry. Bobst Manchester Ltd recommend alloy 1080, which has a purity of 99.80 percent.

However, less pure wire alloys, typically 1100, are extensively used in the metallizing industry.

The alloys that are normally readily available are as follows;

Alloy	1100	1350	1075	1080	1188	1199
% purity	99.00	99.50	99.75	99.80	99.88	99.99

Table 10.4-1 Aluminium wire alloy purity

The wire used in the metallizer will be generally 1.6 mm or 2.0mm diameter. The correct range of diameters that this machine can accept will be listed in the machine specification.

The correct temper of wire is important. All machines manufactured by Bobst Manchester require H18 (USA) or H8 (UK) temper. There is a difference in code used dependent upon the international standard used.

It is essential that the wire used in the machine is clean (i.e.: solvent degreased). Wire is generally purchased on individual spools is supplied clean.

However, some metallizing companies buy wire in bulk and wind it onto spools themselves. It must be highlighted that this can lead to winding quality and cleanliness issues.

Usage of the aluminium wire will depend upon the machine process speed and the thickness of aluminium deposit.

10.5 Lubricants

The amount of different lubricants used on this machine will vary between components and the use of them.

Details are provide elsewhere in this manual of recommendations of how often lubricants should be changed.

Ensure that only the correct approved lubricants recommended for the machine are used.

	Note !
	Use of greases and oils, which are not rated to operate within a vacuum environment, can lead to vacuum process problems which maybe difficult to overcome. The majority of International lubricant companies can offer either the approved or an equivalent.

[See chapter 12.3 Lubrication on page 710 for details.](#)

10.6 Subsystem Antifreeze

The machine currently uses an antifreeze coolant mixture in two sub systems.

Process Chiller Antifreeze

The process chiller unit uses a solution made of a mixture of ethylene glycol, a corrosion inhibitor and water.

	Note ! To minimise the possibility of corrosion issues within the cooled liquid circuits, it is recommended that an approved inhibitor be mixed in the coolant on a 0.5% ratio. Non-use will lead to internal corrosion, and a reduction in the life cycle of the rollers
---	---

The quality of the mono ethylene glycol mixture should comply with that specification described by the chiller manufacturer.

Data sheets for the mixture supplied by the process chiller manufacturer are listed in the relevant section of the supplier's manual

Busch Pumps Antifreeze

The pump uses a medium made of a mixture of ethylene glycol/inhibitor and water for heat transfer purposes.

Glycol (Ethylene), Busch part no 0831 558 702

The mixture is by volume 40% Glycol, 60% water

Amount 212 litres mixture/pump

Data sheets for the mixture supplied by the manufacturer are listed in the relevant section of the supplier's manual.

10.7 Evaporation Section Contact Blocks and Wire Feeder tubes

The following parts used in the evaporation source area should be considered as consumable parts and will need to be replaced on a regular basis. This time scale will depend upon the machines operation and operator cleaning methods.

Contact Blocks

The copper blocks that the evaporators are held between are considered a consumable part.

Please refer to other sections of this manual for information on why they require replacing, under normal machine operation

Material Details

- The copper can be purchased in rectangular lengths
- The copper is described as highly conductive, corrosion resistant and malleable.
- CU 1212.120
- PB 0.005
- Bi 0.0070

Specification

High Density, Highly Conductive (HDHC) Electrical grade copper

Properties

Tensile Strength - 14-25-tons/sq. inch Vickers Hardness - 45-115 VH

Source of Supply

Precious metal dealer example:-

<http://www.anchorbronze.com/c12120.htm>

Live and Neutral Blocks Design Drawings

The blocks are manufactured by Bobst Manchester Ltd., and can be delivered on a relative short delivery.

If you require a drawing of the consumable part for the contract. It can normally be found from within the BOM provided on the electronic documentation CD/DVD

Source Area Compression Seals

When replacing water cooled components around the evaporation source area, it is recommended to replace the seal between the “mating contact surfaces” at the same time.

These seals have been selected for their long operational life under the localised temperature conditions

	Note !
	It has been Bobst Manchester Ltd experience that if the correct seal material and size is not used, problems can occur with water leakage and blockage within this area.

Seal Design Drawing

The seals are manufactured for Bobst Manchester Ltd., and can be delivered on a relative short delivery.

If you require a drawing of this part for the contract. It can normally be found from within the BOM provided on the electronic documentation CD/DVD

Wire Feed Guide Tubes

Bobst Manchester Ltd. will normally supply a machine with copper wire feed tubes. These are the tubes that pass into the source area and are coated with aluminium as it is evaporated.

It is normal to replace these tubes on a regular basis.

Copper Tube

General Details

- Tube wall thickness 1mm
- Tube Outside diameter 6mm
- Tube Inside diameter 4mm

Specification

Half Hard Copper Tube- Allows easy formation of guide tube, but can withstand a reasonable level of heat before distorting.

Source of supply

Enots UK - Pneumatic Distributor part # CS7006003 - <http://www.norgrencatalogue.com/uk/>

Tube Design Drawing

If you require a drawing of the consumable part for the contract. It can normally be found from within the BOM provided on the electronic documentation CD/DVD

	Note ! It has been found that if the tube is not manufactured to this material specification, various process issues can be caused.
--	---

10.8 Process Gases

This chapter describes the process gasses used on the machine.

Plasma Process

Dependent upon the machine specification, the machine will be supplied with a single or multiple gas controllers, called "Mass Flow Controllers" (MFC).

It is very important that the user fully understands, what equipment has been installed on his equipment, before connecting or flowing gas through the gas distribution system.

 	Warning ! All highlighted safety precautions should be adhered to at all times. Failure to follow these recommendations will increase the risk of an explosion
---	--

Gas Purity

The gas purity should normally be 100%.

Guide to Purity Grade Designation

(<http://www.airgas.com/content/details.aspx?id=7000000000012>)

Each product is available in a range of purity grades. These purity grades are identified by names such as "Research Grade", "Ultra High Purity", "Zero", "Pre-purified", etc. that give an indication of relative purity such that "Ultra High Purity" is purer than "Pre-purified". These terms, however, are not absolute. "Research Grade" Argon is not the same purity as "Research Grade" Methane or even "Research Grade" Nitrogen. The same purity name may be used for the same gas by different manufacturers, each offering the product with different specifications. The grade name therefore does not tell the user what the actual purity level is – it varies from product to product and manufacturer to manufacturer.

The important measures of purity are the overall **percentage purity** of the gas and the identity and levels of the **impurities** present.

A simple system used to signify the percentage purity is a digitized grade system. In this system the first digit signifies that number of 9's in the percentage purity. The second digit indicates the next number that differs from a 9.

Therefore for

Helium 4.5

There are four 9's in the purity and the next digit is a 5 –

99995 as a purity percentage is 99.995%

Similarly Argon 6.0 is 99.9999% (Six 9's and the next digit is a 0)

Often of more importance in a particular application is knowing what impurities are present and at what level. If Nitrogen is used as a zero gas for a ppm Oxygen analyzer, it is important to know the Oxygen level of each grade of Nitrogen when selecting which to use.

Each grade of gas will also carry a list of the impurities analyzed for and their maximum permitted levels.

Contact Bobst Manchester Ltd for further advice

[This page intentionally left blank]

10.9 Rolcol Anti Seize Compound

To assist with removal of components on the machine, a anti-seize compound is used on surfaces outside the vacuum chamber. This is copper in colour. Bobst Manchester Ltd, recommend the Rolcol product described on the data sheets below.

Technical Data

ROCOL®

Performance you can trust

ANTI-SEIZE Compound

Copper based anti-seize paste

Product Overview

ROCOL® ANTI-SEIZE Compound is a copper based anti-seize paste reinforced with graphite and molybdenum disulphide to further enhance its performance particularly in applications where conventional copper based anti-seize products may fail to perform.

ROCOL ANTI-SEIZE Compound is designed for use on all static fasteners and mechanisms prone to seizure. This high performance compound is ideal as an assembly and anti-seize lubricant in extreme adverse conditions where pick up and seizure issues may be experienced.

ROCOL ANTI-SEIZE Compound is particularly suited to extreme wet conditions even when submerged in sea water environments.

ROCOL ANTI-SEIZE Compound is also available in aerosol form – see Anti-Seize Spray (ROCOL part code: 14015).

Features and Benefits

- ROCOL ANTI-SEIZE Compound has an excellent static operating temperature resistance of -50°C to +1100°C.
- Non soap thickener producing non melting anti-seize compound.
- ROCOL ANTI-SEIZE Compound prevents pick-up and seizure of static threaded fasteners.
- Lubricates, protects and eases dismantling.
- ROCOL ANTI-SEIZE Compound is effective even in the most aggressive environments and is completely insoluble in water.

Directions for Storage and Use

- Apply as a thin film by brushing or wiping onto a clean dry surface.
- For best results apply to both the male and female parts.
- Also available as ANTI-SEIZE Spray for application by aerosol.
- The storage temperature should be controlled between +1°C and +40°C.
- Shelf life is 5 years from date of manufacture.

Typical Applications

- Furnaces.
- Exhaust systems.
- Dockside and oil rig applications.

Specifications

- Naval Cat No.: 0475-541-8659 (8kg)
- RAF Ref No.: 34D/301-6210 (85kg)
34D/224-6794 (500g)
- NATO Stock No.: 8030-99-301-6210 (85g)
8030-99-224-6794 (500g)
8030-99-541-8659 (8kg)

Pack Sizes

Pack Size	Part Code
85g	14030
500g	14033
6kg	14035
18kg	14038

T +44 (0) 113 232 2600

F +44 (0) 113 232 2740

E customer-service@rocol.com

www.rocol.com

ROCOL House, Swillington, Leeds LS26 8BS

Registered Company No. 559693 VAT No. 742 0531 67

Registered Office: Admiral House, St Leonard's Road, Windsor, Berkshire SL4 3BL

 ROCOL A division of **ITW** Ltd

 BS EN ISO 9001
Certificate No. 041 12448

 BS EN ISO 14001
Certificate No. 045 67556

 OHSAS 18001
Certificate No. 043 38129

 INVESTORS
IN PEOPLE

Technical Data

Performance you can trust



ANTI-SEIZE Compound

Copper based anti-seize paste

Property	Test Method	Result
Appearance	N/A	Dark coppery coloured paste
NLGI No.	IP 50 – ASTM D217	1/2
Base Type	N/A	Mineral oil
Thickener	N/A	Organically modified clay
Solids	N/A	Copper, Graphite, MoS ₂
Solids Content	N/A	Approximately 37%
Temperature Range	N/A	-50°C to +1100°C
Water Solubility	N/A	Insoluble
Coefficient of Friction	1.25" carbon steel bolts	0.15
Approximate Coverage	0.1mm film thickness	10m ² /kg

Values quoted above are typical and do not constitute a specification.

Safety Data Sheets

Safety data sheets are available for download from our website www.rolcol.com or may be obtained from your usual ROCOL contact.

The information in this publication is based on our experience and reports from customers. There are many factors outside our control or knowledge which affect the use and performance of our products, for which reason it is given without responsibility.

Issue: 2 Date: 02-13

T +44 (0) 113 232 2600

F +44 (0) 113 232 2740

E customer-service@rolcol.com

www.rolcol.com

ROCOL House, Swillington, Leeds LS26 8BS

Registered Company No. 559893 VAT No. 742 0531 47

Registered Office: Admiral House, St Leonard's Road, Windsor, Berkshire SL4 3BL

ROCOL A division of **ITW** Ltd



BSI 02 3001
Certificate No. 44 12448



BSI 01 14001
Certificate No. 05 67596



BSI 01 18001
Certificate No. 04 781 721



INVESTORS
IN PEOPLE

[This page intentionally left blank]

SECTION 11.0 MACHINE CALIBRATION PROCEDURES

This section of the manual describes how components and systems supplied on the machine should be calibrated under normal machine operation

- [11.1 ABB Load Cell System Calibration on page 521](#)
- [11.2 Optical In-Line Monitor Probe Calibration on page 526](#)
- [11.3 Vacuum Gauge Maintenance on page 532](#)
- [11.4 Auto-loss Sequence \(Winding Mechanism\) on page 539](#)
- [11.5 Rewind Layarm Alignment on page 542](#)
- [11.6 Motorised Spreader Roller Replacement & Position Feedback on page 549](#)
- [11.7 Unwind Layarm Roller Mechanical Alignment on page 552](#)

	Note !
	Some of the equipment listed in this section of the manual may not be applicable to this machine.

11.1 ABB Load Cell System Calibration

The load cell system consists of number sets of load cell transducers, cables, and leadthroughs and amplifier cards. The transducers are mounted in pairs at each end of a lightweight roller. They are wired via a vacuum leadthrough to the amplifiers, which are located on the faceplate, on the outside of the chamber.



Figure 11.1-1 Load cell transducers in a machine



Figure 11.1-2 Load cell amplifiers

Load Cell System Adjustments

Load cells and their associated amplifiers normally remain in calibration for long periods of time. However, should the process roll condition harden or soften more than expected, whilst running at the normal tension settings, it is advisable to check the load cell system calibration.

The output signal is fed to the drive system to provide indication of the tension on the substrate and also to control the unwind and rewind drives.

Calibration

The calibration of the system generally follows the following procedure.

Choose either the unwind or rewind load cell system. The amplifiers are identified.

These instructions are a general guide and the manual supplied by the load cell manufacturer should be referred to before and during the calibration procedure.

Using Fast Set Up Feature for recalibration



Steps back to previous menu.

Sometimes you must press this button two or more times to get back to the desired menu



Steps up in a list.



Steps down in a list.

Steps to the next main menu



OK (acknowledgement) button. Confirms a selection or a parameter setting

Table 11.1-1 ABB Load cell amplifier interface panel

Locate the amplifiers on the winding cart. [See Figure 11.1-2 on the previous page for details.](#)

Performing a Fast Setup

The fast setup guides you through a minimum number of steps to set up the tension electronics.

You are asked to answer some questions and to enter desired values. These selections and parameter settings must be made to get the tension electronics ready for measurement.

Only a limited number of selections and parameter settings are set in the fast setup. All other parameters are set as factory default values.

The fast setup can be made in two ways, depending on how the wrap gain is set.

The wrap gain can be set by selecting “HangWeight” or “EnterWrapGain”. In our case we always use “Using Hanging Weights” method.

The simplest method to set the wrap gain in this tension system is to use a known weight that loads the centre of the roll with a rope that follows the web path exactly. When going over rubber rollers, the load should be spread using a piece of cardboard or plastic to spread the load and not to cut into the roller.



Caution !

When using heavy weights, take necessary precautions to support them and to minimise the possibility of damage, if they are dropped.

Below is the instruction set from the vendor’s manual on setup.

For recalibration, all the following underlined values are already known to the system. And only need to be confirmed.

Step	Pictorial	Instruction
1		Press for 5 seconds to go to the FastSetup menu
2		Press to start the fast set-up sequence
3		Select Hanging Weights from the list with up and down cursor. Confirm with
4		Select Two Cells , (at this stage, just press the Ok button)
5		As per the transducers information, (at this stage, just press the Ok button, as it should have been set up by the commissioning engineer.)
6		With only the dead weight load of the roller, carry this out. No weights applied

Thread a flat tape around the rolls so that a spring balance or weight can be used to apply load in the same manner as the substrate. Ensure the tape is in the middle of the roller. [See Figure 11.1-3 on the next page for details](#) and [Figure 11.1-4 Rewind load cell calibration webpath on the next page](#)

7		The value being applied by the weights should be entered here as a value in newtons. Confirm with 
---	---	--

	Note !
	Do not hang the weight/belt over the unwind or rewind shafts, as they are driven and will have frictional losses. This will affect the calibration.

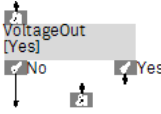
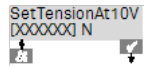
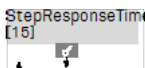
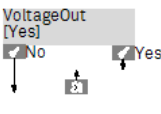
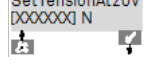
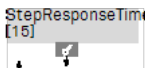
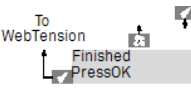
8		Select Voltage and press  to set up the voltage output.
9		The full range value as per the machine specification in section 15.1 of this manual, multiplied by 1.1 and then entered. Confirm with a 
10		Leave at 15 mS Confirm with a 
11		Select Yes Confirm with a 
12		The full range value as per the machine specification in section 15.1 of this manual, multiplied by 1.1 and then entered. Confirm with a 
13		Leave at 15 mS Confirm with a 
14		Press  to finish the fast setup and go to the normal view display

Table 11.1-2 Performing a fast setup procedure for ABB load cell system

In the case of a machine that has a different wrap angle for under/overwind web paths on the unwind, always use the unwind path.

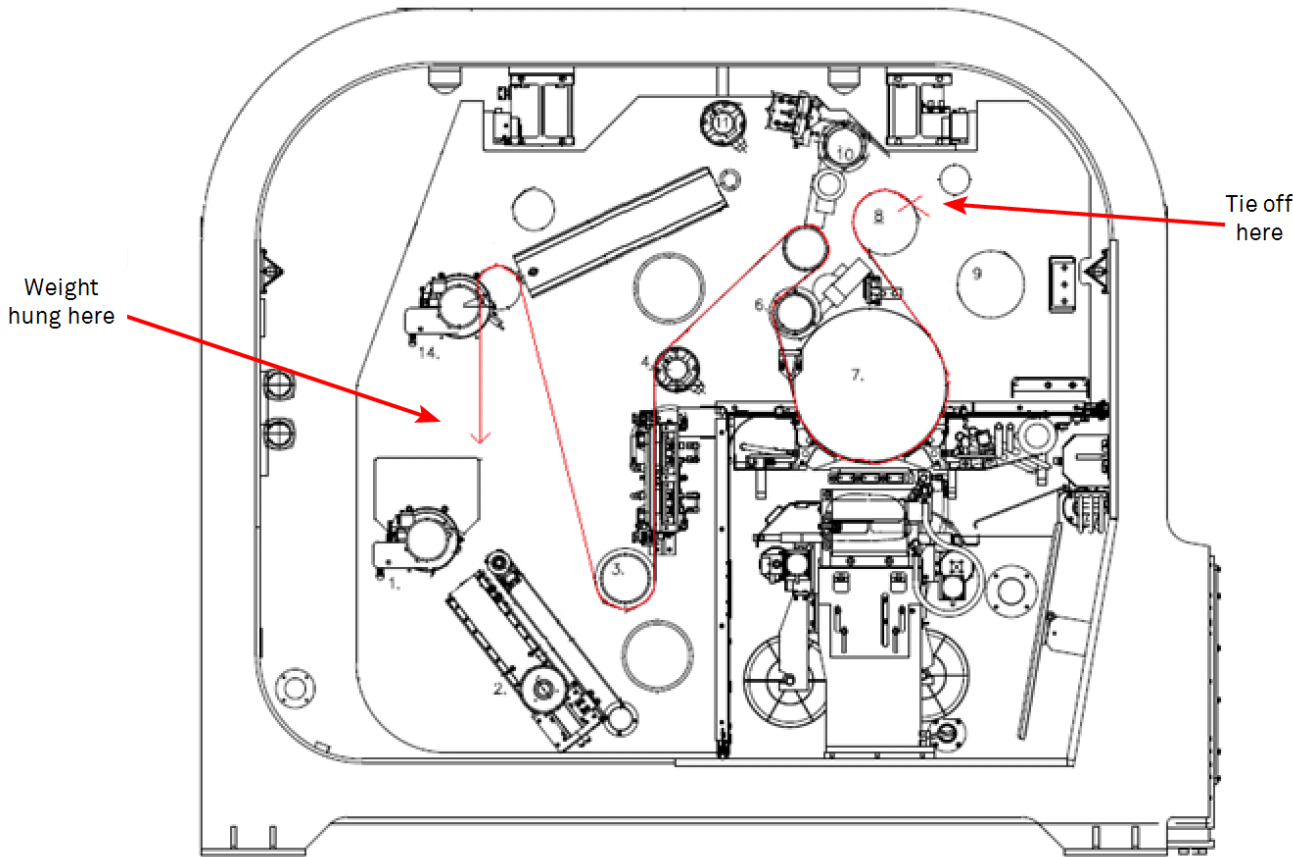


Figure 11.1-3 Unwind load cell calibration webpath

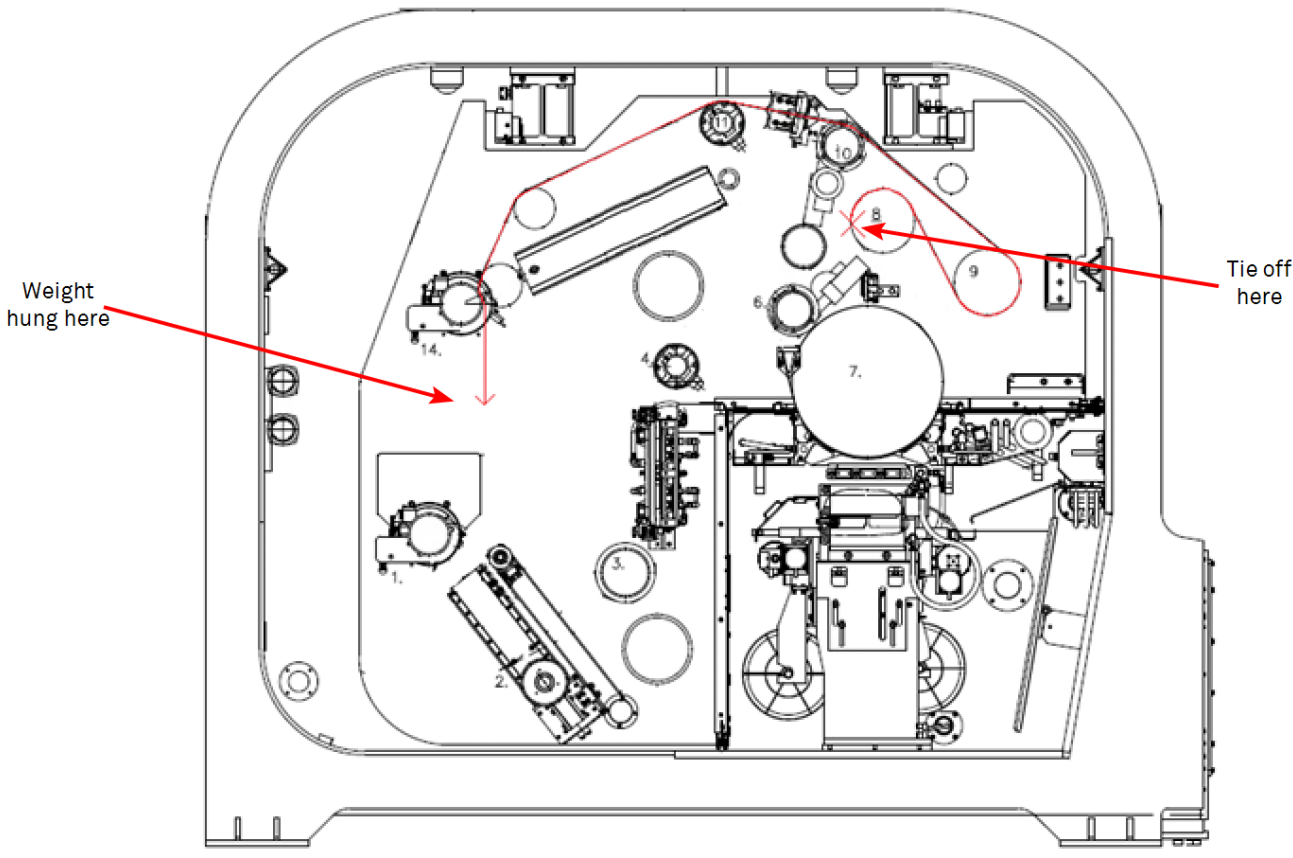


Figure 11.1-4 Rewind load cell calibration webpath

11.2 Optical In-Line Monitor Probe Calibration

There is a requirement to calibrate the non-contact monitor system in the machine. Most end users carry out this task at different periods to comply with their quality systems.

The following procedure is a general guide.

Equipment Required

A calibration tool is provided to allow the calibration of the optical monitor system.

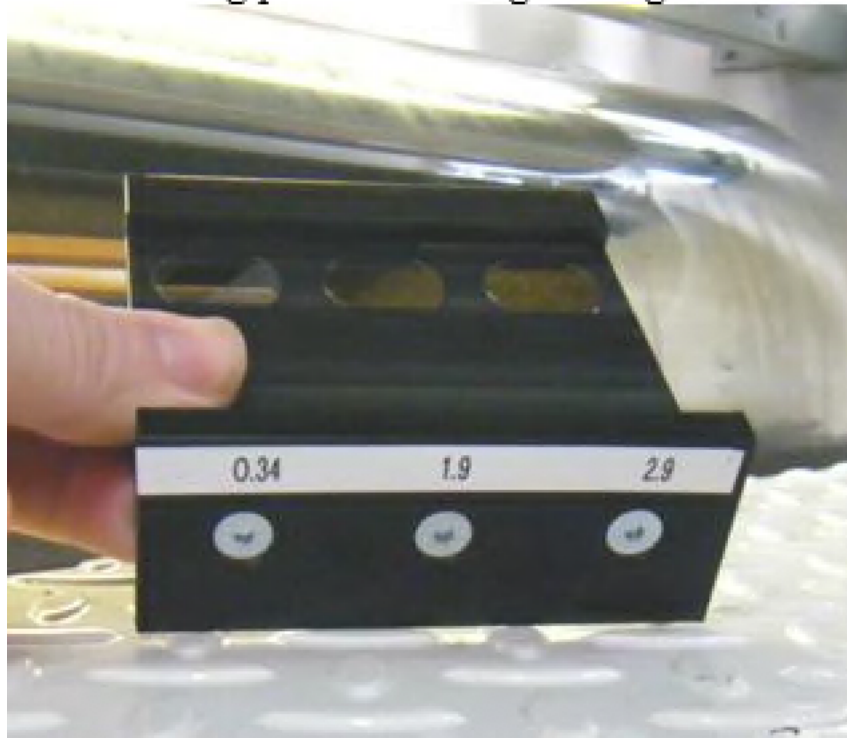
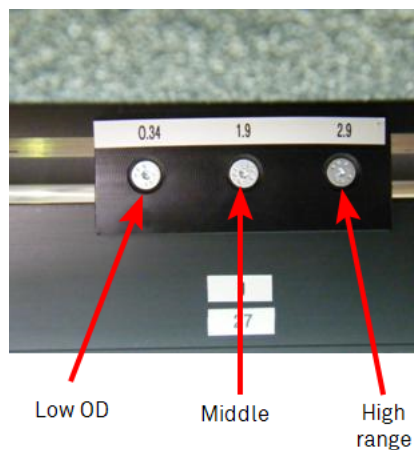


Figure 11.2-1 Calibration tool

Choosing a calibration target

There are three options on the calibration tool

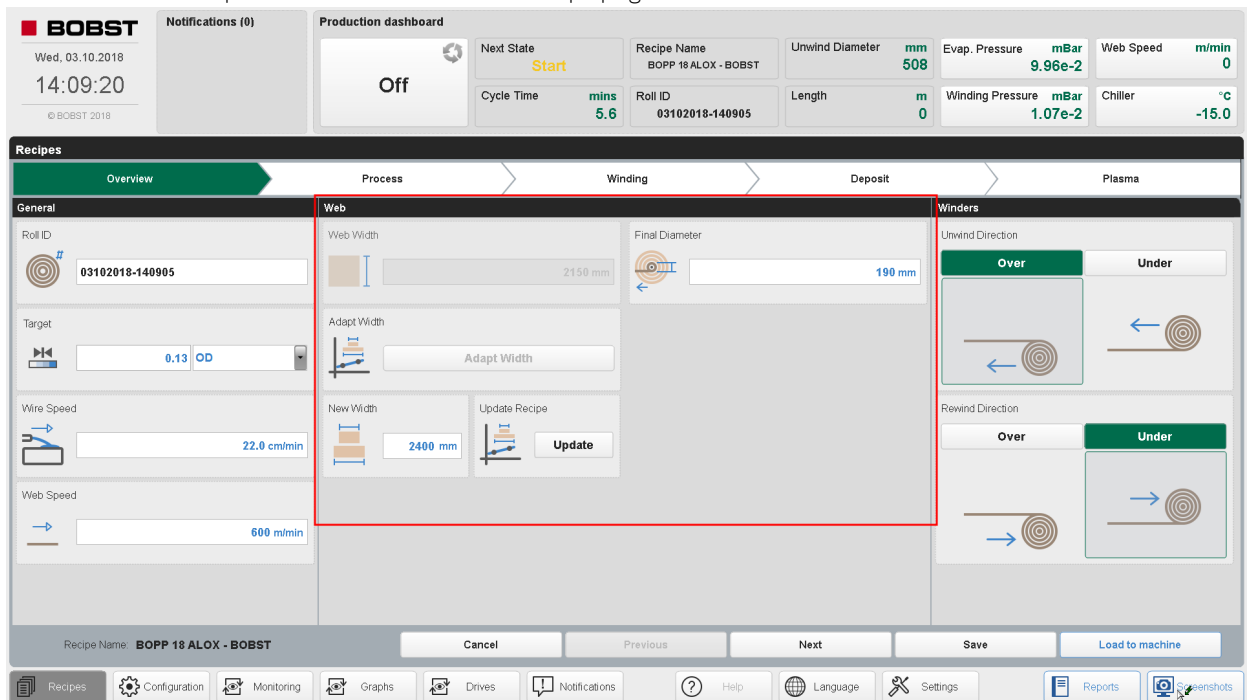


- Use the low option on the tool when calibrating the probes for low OD work (OD values below 0.5)
- Use the middle option on the tool when calibrating the probes for general products. It may also be used when calibrating the high and low OD ranges.
- Use the high option when you are only calibrating the high OD range or when calibrating the monitor hardware.

	Note !
	It is possible to calibrate both high and low ranges with the same calibration sample, as long as it is between the range 1 OD and 2.5 OD. The recommended calibration sample value is 2 OD.
	If a calibration sample outside this range is to be used, each calibration has to be selected for the purpose of calibration.
	Low range 0.2 > 2.0D High range 1.9 > Higher

Calibration Procedure

1. Switch the machine power supply on.
2. Allow 30 minutes for the machine to warm up
3. Ensure that all web has been removed from the machine
4. Go to the HMI and open the Overview tab in the Recipe page:



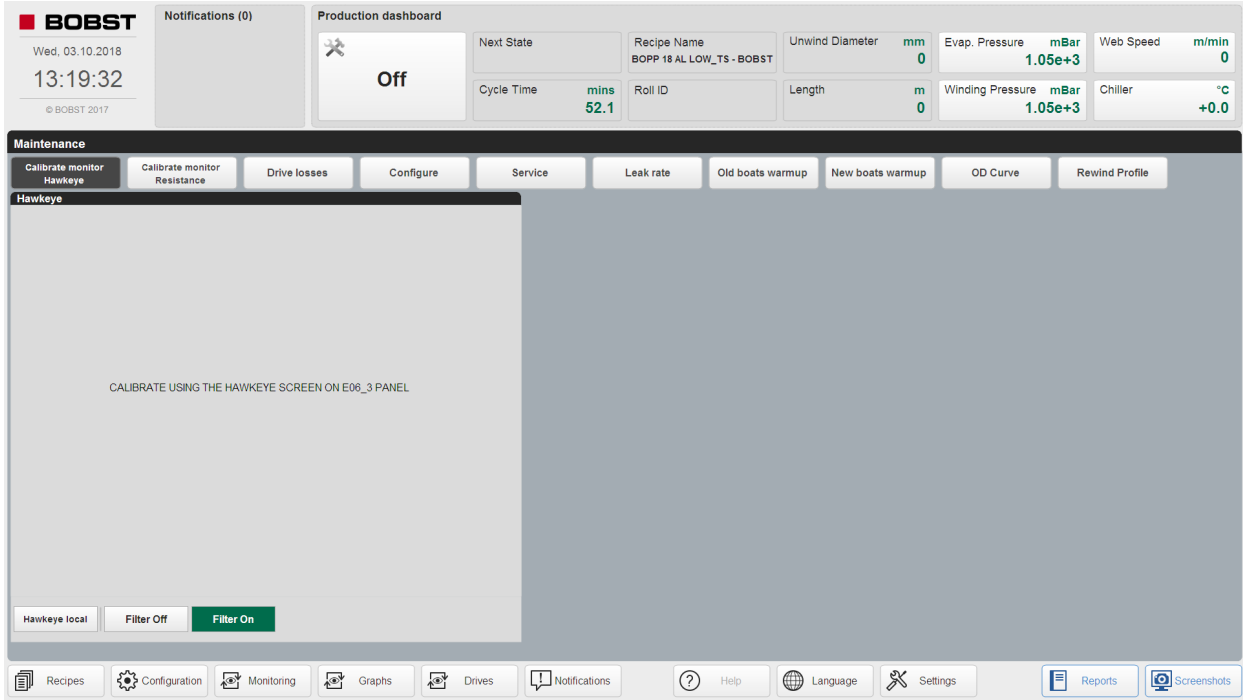
5. Enter the width value in the appropriate field:
 - Enter the maximum width in the new width field if the adapt width feature is enabled.
 - Enter the maximum width in the web width field if the adapt width feature is not enabled.
6. Open the Deposit tab and select the Deposit 2 setting.

The screenshot shows the BOBST Production dashboard. The top bar includes the BOBST logo, date/time (Wed, 03.10.2018 14:11:20), and a production status of 'Off'. The main area is divided into 'Recipies' and 'Production dashboard'. The 'Recipies' section has tabs for Overview, Process, Winding, Deposit (selected), and Plasma. Under the 'Deposit' tab, the 'Setting Type' is 'Deposit 1'. The 'Curves and Monitor Control' section shows 'Default_OD' and a 'Monitor Control' area with 'Hawkeye' and 'Resistance' options. The 'Pinholes' section shows detection zones and warning levels. The bottom bar includes buttons for Cancel, Previous, Next, Save, and Load to machine.

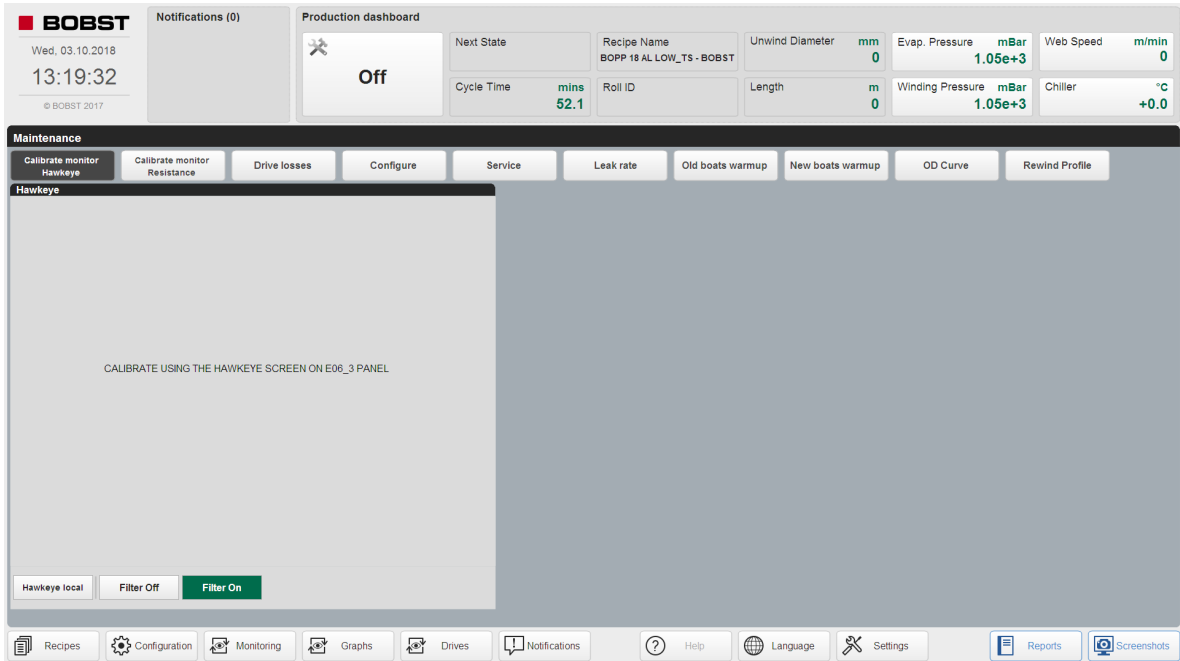
7. Set the Monitor Control function to OD
8. Select the Deposit 1 setting:

The screenshot shows the BOBST Production dashboard. The top bar includes the BOBST logo, date/time (Wed, 03.10.2018 14:11:17), and a production status of 'Off'. The main area is divided into 'Recipies' and 'Production dashboard'. The 'Recipies' section has tabs for Overview, Process, Winding, Deposit (selected), and Plasma. Under the 'Deposit' tab, the 'Setting Type' is 'Deposit 1'. The 'General' section shows 'Target' set to 0.13 OD, 'Pre Gas Target' set to 0.40 OD, and 'Web Transmission' set to 99.9 %T. The 'Deposit Alarms' section shows various alarm levels. The 'Offsetting' section shows 'Offsetting' set to On, 'Low Limit' set to 75.0 %T, and 'High Limit' set to 99.9 %T. The bottom bar includes buttons for Cancel, Previous, Next, Save, and Load to machine.

9. Set Target unit to OD and set the Target OD value
10. Set the Web Transmission value to 99.9%
11. Press Load to Machine
12. Go to the Setings menu and open the Maintenance page. Select the Calibrate Monitor OD settings:



13. Press turn off red-shift
14. Press turn filter off
15. Press selftest. This will check to see that there is no material in front of the probes. Any faults will be identified.
 - A pop up message will appear if the votage offset is changed by more than 0.1 from the previous value. It will suggest that there is web in the machine and ask if you want to force or bypass. Select force if there is no web in the machine



Channels/probes are numbered by boat number. Details of each probe are displayed in the table. The nominal values for each parameter reflect the values that should be showing for a new monitor.

Parameter	Nominal Value	Tolerance
Gain	1000	+/- 100
Voltage	0.2	+/- 0.1
OD Offset	1	+/- 0.1

If the values fall outside of the acceptable tolerances, follow the cleaning and maintenance procedure.



Figure 11.2-2 Monitor beam system

16. Line sample up with the appropriate window. Please refer to the table above for guidance on which sample to use



Figure 11.2-3 Example of a technician inserting the calibration jig

17. Place the jig in front of the probe so that the middle line on the jig can be aligned with the mark on the beam. It is important that the jig is placed in the same position in front of every probe, with alignment to the nominated calibration optical lens.

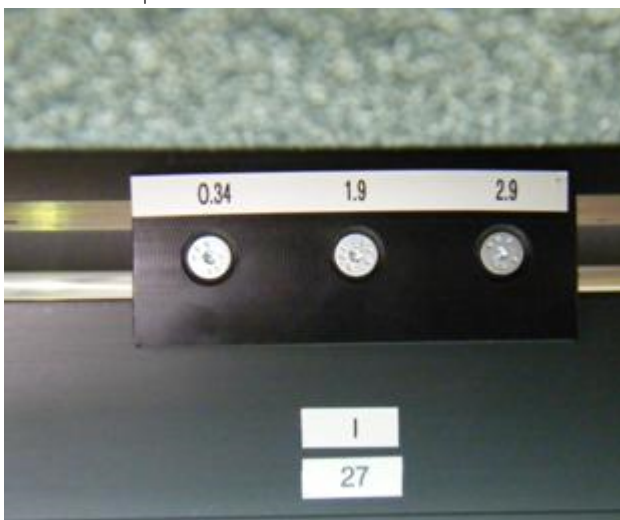


Figure 11.2-4 Example of a technician aligning the calibration jig with the receiver probe.

18. Start at probe number 2 on the left-hand side. Use the calibration tool to calibrate each of the probes on the monitor beam:
 1. Select channel 2 on the probe voltage graph.
 2. Press Calibrate if you are calibrating the low or middle OD ranges
 - Press the calibrate high option as well if you are also calibrating the high OD rangeThe key will show amber when the calibration is running. It will turn green or red when the calibration is complete.
 - The key will turn green when the calibration is complete
 - The key will turn red if the calibration fails
 3. Verify that the OD value on that channel has met the calibration OD value.
 4. Repeat steps 16 to 18 for the next channel. Repeat until all probes have been calibrated and all probe channels are green.
19. Perform spot check on a few probes at random.
20. Press save/calibrate.
21. Turn filter on
22. Turn red shift on
23. Re-load correct recipe before running the production.

11.3 Vacuum Gauge Maintenance

This chapter describes how to maintain the vacuum gauges on the machine.

Dial Gauge

This type of mechanical dial gauges require little maintenance and should only be cleaned as required.

Calibration

Calibration of the instrument must be undertaken only by the manufacturer and therefore should be returned to Bobst Manchester Ltd.

Edwards Active Inverted Magnetron Gauge

This gauge is only operational when the high vacuum valve (VV7) is opened.

The gauge is only used for monitoring purposes of the vacuum level in the evaporation zone. It is not used for any interlock circuits.

If the vacuum reading from the Active Inverted Magnetron Gauge head is erratic, the problem is usually due to contamination. To clean the gauge head, follow the instructions in the vendor manual supplied under section 5.5.



Figure 11.3-1 Active inverted magnetron gauge

MKS Pirani Gauge

If the vacuum pressure reading from the Pirani gauge is erratic on the HMI screen, a possible reason is that the gauge needs cleaning or is out of calibration.



Figure 11.3-2 MKS pirani gauge

Cleaning, Preventative Maintenance

The maintenance and calibration of the gauge is described in detail within the manufacture's manual. This is a practical guide of the calibration on the machine.

	Warning ! The manufacturer advises, “If the transducer is backfilled with a liquid like pump oil the sensor element is likely permanently damaged. The transducer cannot be cleaned using solvents.”
---	---

Calibration

To carry out the calibration at the lower vacuum end of the gauges measurement range (zero), it is suggested that the gauge be placed within a diffusion pump at the gauge point (PIR 1, 2 or 3 position).



Figure 11.3-3 MKS pirani gauge positioned in the valve area (pod) above the diffusion pump.

To carry out the calibration, the following procedure should be followed:-

**Warning !**

Do not vent the diffusion pump to atmosphere, as a potential explosion hazard can occur.

1. The diffusion pump should have been in operation for at least 1 hour.
2. Gain access to the MKS pirani gauge location at the diffusion pump by removing the access cover.
3. Close the hand valve fully between the pump and the existing pirani gauge (located in the pump's inlet)



Figure 11.3-4 Example of a technician isolating the Pirani gauge by closing the hand valve.

4. Remove the electrical connection from the gauge carefully
5. The power to the pump will be switched off at this time automatically, as it senses a loss of connection.
6. Remove the gauge and replace it with the gauge to be calibrated.
7. Reconnect the electrical connection back into the gauge.
8. Allow 5 minutes for the gauge to heat up the internal electronics.

Setting Atmospheric Pressure

9. Press down the user button for 2 seconds and the LED will flash green three times to acknowledge the atmospheric adjustment has been executed successfully. The LED will flash red three times if the adjustment failed.

**Note !**

Atmospheric adjustment only changes the high measuring range and will have no influence on measuring errors in the range below 13 mbar.

Setting ZERO Setting (low vacuum)

1. Open the valve between the gauge and pump.
2. Ensure that the pump is switched back on from the vacuum HMI screen (maintenance access level) and allow for a minimum of 20 minutes for the correct vacuum conditions to be established.
3. The vacuum pressure level within the diffusion pump should be less than $1e^{-5}$ mbar after this period, assuming that the pump is in a relatively clean condition and a full charge of oil is present.
4. Monitor at the voltage being feedback on the VACUUM HMI screen, monitor that the volts feedback for the gauge point that the gauge has been installed in.
5. When the voltage feedback is less than 1 volt dc, the ZERO button can be pressed on the gauge to carry out the "ZERO" calibration.
6. Press down the user button for 2 seconds and the LED will flash green three times to acknowledge the atmospheric adjustment has been executed successfully. The LED will flash red three times if the adjustment failed.
7. After carrying out the calibration procedure, close the hand-valve and replace the calibrated gauge with the original unit.

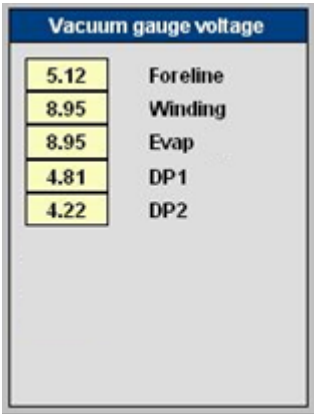


Figure 11.3-5 Screen capture of the “Vacuum” HMI screen showing the Pirani gauge feedback signals



Figure 11.3-6 Pirani gauge showing the status LED and calibration button.

	Note !
	If the diffusion pump has not been allowed to heat up, to its operating temperature the gauge may be incorrectly calibrated, which will lead to problems in the pump's operation. This is because the computer system recognises that if the voltage signal falls to below 0.75 Vdc, it assumes that the electrical communications lead has been removed and the pump will be switched off.

Vacuum Gauge - Pressure Level Settings/Interlock Activation

The MKS pirani gauges provides 1-9 DC volt signals to the Siemens PLC analogue input modules that are relative to the vacuum level at the point in the system that the gauge is mounted.

In the machine’s software, different switch points for interlocking of the machine based upon the gauge feedback signals.

	Warning !
	The interlock switch settings should not be altered other than by the Bobst Manchester Ltd. Engineer, as the values set comply with component vendors requests.

The machine compares the incoming signals to the pre-set levels and when they are the same, it will send a signal to allow the automated interlocked operation to take place.

The software has been designed to have a differential switch function.

If the cable between the gauge and the PLC module is removed, the pressure switch will fail to an open situation and a reading of <1e⁻⁵ mbar or lower would normally be seen on the VACUUM screen.

The purpose of the Inbuilt baffle segments in gauge type MKS 925-81313-0094

The baffle arrangement has four segments with a small hole in each of them to allow the air molecules to be removed, but provide a obstruction against the ingress of the process debris.

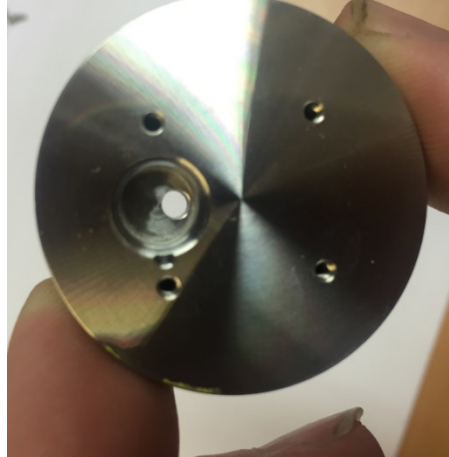


Figure 11.3-7 Example of an existing gauge showing the hole from vacuum system straight to the sensor.



Figure 11.3-8 The baffle segment arrangement in the new gauge.

Cleaning Inside the Baffle Segments

This procedure should be carried out in a clean area to stop further contamination occurring.

It is also recommended that suitable gloves be used to stop potential contamination.

A pair of “internal circlip pliers will be required. These need to be cleaned with acetone to stop contamination.

1. Remove the gauge from the machine to a clean area.
2. Use the circlip pliers to remove the internal circlip and place it in a safe location.



Figure 11.3-9 Example of a retaining circlip removed

3. Lift the gauge upwards and allow the internal baffle segments to slide out of it. Note the order for re-assembly purposes.

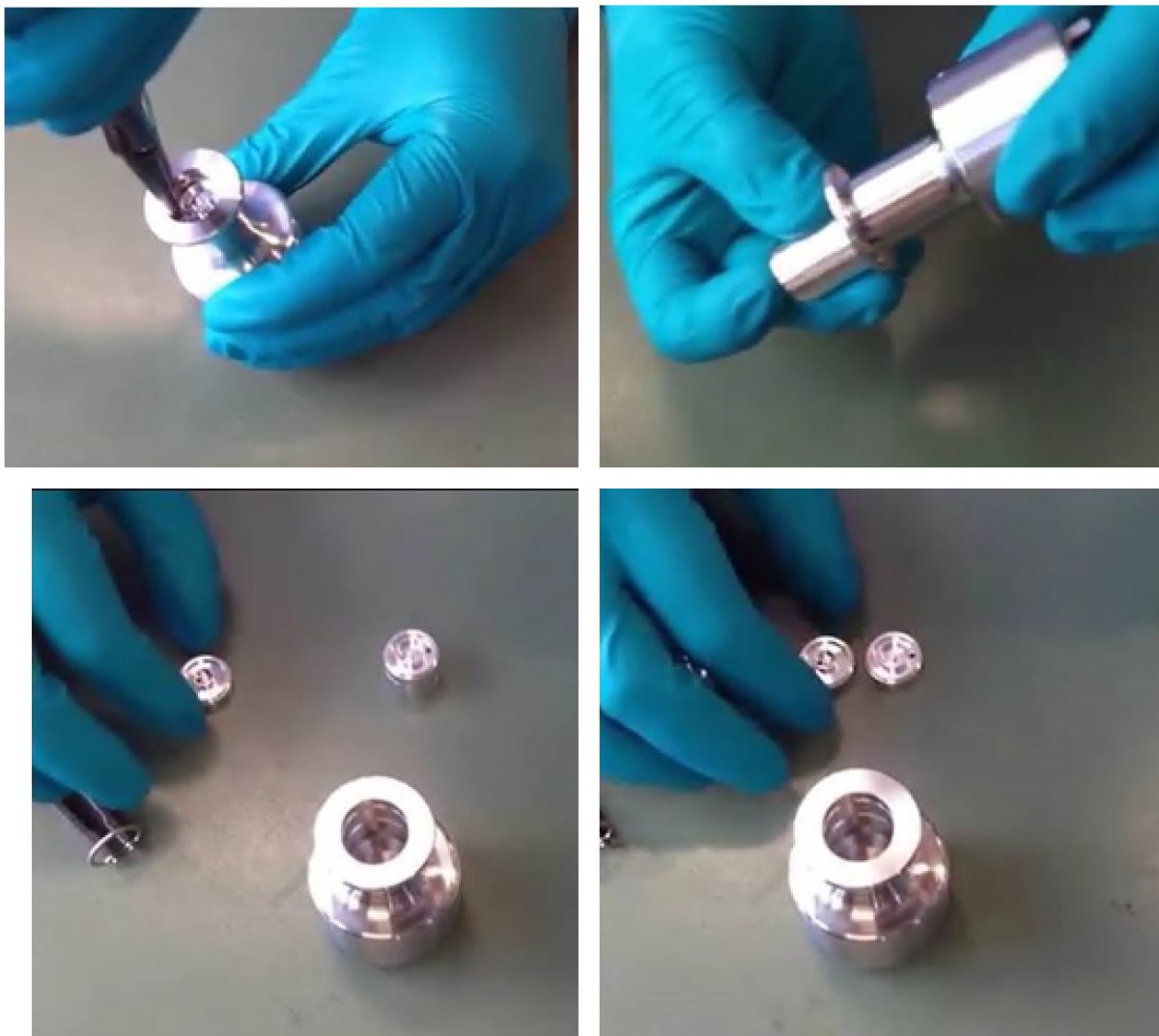


Figure 11.3-10 Different stages of the disassembly

4. Clean each baffle segment with acetone, and allow them to dry.
5. Re-insert the baffles back into the gauge body in the order they were removed.
6. Re-insert the circlip to secure the baffle segments.

Recalibrate the gauge as described earlier.

Location of Vacuum Gauges on Machine

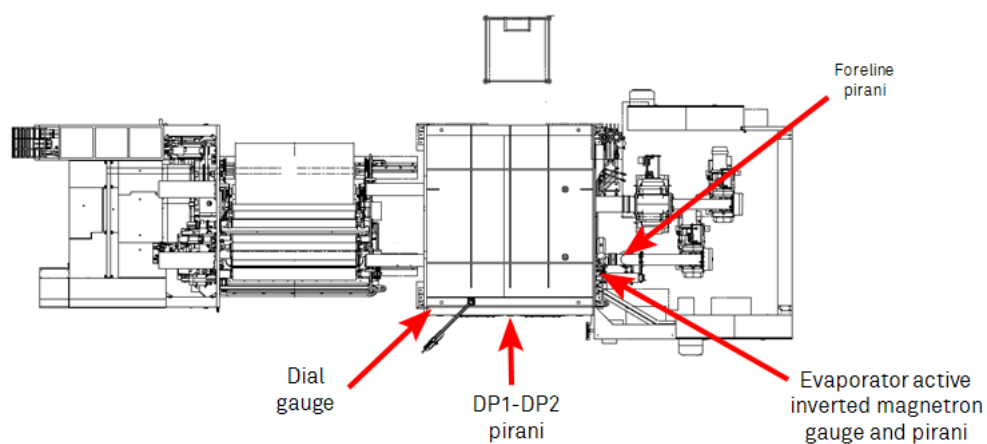


Figure 11.3-11 Plan view of machine indicating location of vacuum gauges

11.4 Auto-loss Sequence (Winding Mechanism)

The following motors can be run in “Torque Control” on the K5 Expert machine range.

- Draw
- Drum
- DTR1
- DTR2

As part of the machine start-up procedures, the frictional losses are taken in to consideration by running a sequence called “Auto-losses”. This sequence will allow the machine to run these rollers over a speed range and monitor the amount of torque required to rotate them at these speeds. These values are recorded in the drive plc and are then used under the normal running conditions of the machine.

Normally, unless there are mechanical components changed, there should be no need to run this procedure.

In the case of process problems where there appears to be issues where the rollers do not reach the torque set-point, it should be considered to run this sequence.



Note !

For this sequence, no substrate should be in the web path.

The machine requires reel shafts placing in both sets of chucks, and where appropriate, chuck locked switches have to be engaged.

Procedure:-

1. The ‘Get diameter’ procedure has to be been completed and then move the layon assemblies back to the home switches, using the local rear push buttons.
2. Place the machine in the normal vacuum mode and allow it to be evacuated to its normal operating pressure, before starting the process.
3. Select the Calibration screen from the supervisor level and then the ‘DRIVES LOSSES’ screen on the machine HMI interface screen.



Warning !

Ensure that the machine is in a safe condition to allow movement to take place.

			Time left	Torque	Spd tol	Sequence
Unwind	Start	Abort	0:00	0.0		Auto loss tuning complete
Rewind	Start	Abort	0:00	0.0		Auto loss tuning complete
Drum	Start	Abort	0:00	0.0		Auto loss tuning complete
DTR1	Start	Abort	0:00	0.0		Auto loss tuning complete
DTR2	Start	Abort	0:00	0.0		Auto loss tuning complete
Draw Roller	Start	Abort	0:00	0.0		Auto loss tuning complete
Abort All			Abort Warmup		Acknowledge Finished	
					Reset Drives	

Figure 11.4-1 Auto-loss HMI Activation Section

4. Start the sequence by pressing the Start button for the drives that the sequence is being carried out on. The drive/motor/roller will then progress to run for a set period in a “warm up mode”.
5. After completing the ‘warm up’, the drive will be run at set speeds and the losses averaged out over the time period. The unwind and rewind drives will be rotated in both a forwards and reverse direction, while the other drives will

only be driven in a forward direction. This is because the winders can be operated in both 'over and under' wind directions.

6. After all the procedures have been carried out, the 'Acknowledge 'Finished'' button must be pressed

	Note !
	It is not essential to carry out the auto-losses for the unwind & rewind drives, as they controlled from load cell feedback and are not run in "Torque mode of
	Note !
	If any fault codes or alarms are indicated though the sequence, investigation must be made to understand why and overcome them

Button/parameter	Description
Unwind/ Draw/DTR1/DTR2/Drum / Rewind	Identified the drive/motors that the auto-losses can be carried out on.
Start	This button starts the auto-loss sequence on the drive/motor selected
Abort	This button aborts the auto-loss sequence on the drive/motor selected
Time Left	Indicates how long the step indicated by the sequence description has left to count down in seconds to zero.
Torque	Indicates the torque being measured while any steps of the sequence are being carried out
Speed Tolerance	Indicates if the actual speed has reached the set speed at any step in the sequence
Sequence	The sequence section of the screen will indicate the stage the auto-loss sequence is currently at. It can also indicate error messages.
Abort All	This button will abort any auto-loss sequences taking place
Abort Warmup	This button will stop the warm up steps of the sequence for all motors/drives selected
Acknowledge Finished	At the end of the sequences when all motor/drives have passed through the steps, this button has to be pressed to acknowledge the completion.

Table 11.4-1 Screen functions Key

Sequence Message	Description
Idle	Sequence not started and system in an idle condition
Warm up in progress	Drive/motor in auto warm up sequence to allow torque to be measured when mechanical components are at temperature
Compensation Speed 1 – active forward	Indicates step in the sequence. There are steps 1-10, which are different speeds.

Table 11.4-2 Sequence Message Descriptions